

STATISTICAL MACHINE LEARNING (WS2025/26)

HOMEWORK: IMAGE RECONSTRUCTION



Assignment (5 points) Detective Cassia Ren, special investigator for the Jedi Archives Division, stood in the quiet glow of the analysis chamber. Before her, dozens of shimmering holo-images flickered—warped, shifted, and drowned in noise, as if the Force itself struggled to hold them together.

“These fragments are all we recovered,” said Master Kel Doril, the Archivist. He adjusted his mask and leaned over the console. “The final transmission from Master Yoda’s hidden holocron—sent moments before the Empire seized the temple.”

Cassia narrowed her eyes. “And this holocron... it contains a secret?”

Kel nodded gravely. “Not just any secret. It carries the name of the only thing capable of stabilizing the hyperspace pathways. Without it, the galaxy will fracture into isolated systems. No travel. No trade. No alliances. Only unending war.”

Cassia turned her gaze back to the corrupted images. Each was a warped, column-shifted version of the same underlying picture—scrambled by a failing transmitter and distorted by turbulence in hyperspace.

“So the holocron tried to send a single image,” she said. “But all we received were these distorted echoes.”

“Precisely,” Kel replied. “We believe each frame is a shifted and noise-corrupted version of the true image. The original message is still there—just deeply buried.”

Cassia folded her arms. “If I’m not mistaken, only apprentices who have mastered the secrets of machine learning can help us now.” Kel nodded in solemn agreement.

Model description: You are given a sequence of m images of size $\text{width} \times \text{height}$, i.e.,

$$(I_i \in (0, 1)^{\text{width} \times \text{height}} \mid i = 1, \dots, m),$$

which are assumed to be i.i.d. samples from a mixture distribution

$$p(I) = \sum_{k \in \mathcal{K}} p(k) p(I \mid k), \quad p(I \mid k) = \prod_{x=1}^{\text{width}} \prod_{y=1}^{\text{height}} \mathcal{N}(I_{x,y}; r(T, k)_{x,y}, \sigma^2).$$

where:

- $I_{x,y}$ denotes the pixel value at position (x, y) in image $I \in (0, 1)^{\text{width} \times \text{height}}$.
- $T \in (0, 1)^{\text{width} \times \text{height}}$ is an unknown clean image that we want to reconstruct.
- $r(T, k)$ is the image obtained from T by shifting its columns by $|k|$ pixels:
 - if $k < 0$, the columns are shifted to the left,
 - if $k > 0$, the columns are shifted to the right.
- $p(k)$ is the prior probability of the shift k .
- $p(I \mid k)$ is the likelihood of observing image I if the true image is T shifted by k and then corrupted by Gaussian noise.
- $\mathcal{N}(z; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(z - \mu)^2\right)$ denotes the univariate normal density with mean μ and variance σ^2 .

Task: Your goal is to reconstruct the unknown image T from its noisy, randomly shifted versions $(I_1, \dots, I_m)$. To accomplish this, you are asked to implement the Expectation–Maximization (EM) algorithm for the probabilistic model described above. In particular, your implementation should:

1. Estimate the posterior responsibilities $p(k \mid I_i)$ for each image and shift k (E-step).
2. Update the estimate of the unknown clean image T and the variance σ^2 using these responsibilities (M-step).